Intron content, a duplicated uncharacterised ORF, and mito-nuclear balance: a comparative and functional anatomy of the *Pleurotus ostreatus* mitochondrial genome

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Abstract

Fungal mitochondrial genomes vary several-fold in size within a single genus, and the usual explanation is intron content. We test that directly across twelve complete *Pleurotus* mitogenomes, annotating all of them with one pipeline rather than relying on their deposited features, which vary enormously in completeness. Size spans 60,694 to 102,950 bp and correlates with detected group I intron content ($\boldsymbol{r = +0.89}$) and more strongly with open reading frame count ($\boldsymbol{r = +0.96}$) — consistent with introns and the homing endonucleases they encode being acquired together. Within *P. ostreatus* alone the range is 65,277 to 73,085 bp, so the same process operates below the species level. The blue oyster mitogenome contains little internal duplication: two repeats totalling 802 bp (1.1% of the genome), but one of them duplicates an uncharacterised open reading frame at 84.5% identity. Of twenty predicted ORFs with no Swiss-Prot homologue, a more sensitive Pfam search assigns function to none, yet four are conserved in at least six of eleven other *Pleurotus* mitogenomes and expressed in at least half of sixteen RNA-seq libraries — candidate genes that current databases cannot name. Screening the nuclear proteome against MitoCarta3.0 and curated Swiss-Prot localisations predicts 631 nuclear-encoded mitochondrial proteins (5.0% of the proteome), and the ratio of mtDNA-encoded to nuclear-encoded mitochondrial expression is highest in the exudophore, though replicate ranges overlap and this remains an observation rather than a result.

Keywords: mitogenome, *Pleurotus ostreatus*, group I intron, gene duplication, MitoCarta, mito-nuclear communication

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Companion to the mitogenome annotation paper, which describes the corrected annotation and the intron evidence in full. All values are generated from the analysis repository and verified by `scripts/32_audit_numbers.py`. [**AUTHORS**] markers indicate items only the authors can supply.

1 Introduction

Fungal mitochondrial genomes are unusually plastic. Within a single genus they routinely vary several-fold in length while encoding essentially the same small set of proteins, and the standard explanation is mobile intron content: group I introns, and the homing endonucleases they carry, invade conserved genes and expand the genome without adding respiratory function [1].

That explanation is widely repeated and less often tested within a genus with a consistent annotation. The difficulty is that deposited mitogenome annotations differ enormously in completeness — the companion paper to this one shows that the reference used here carried thirteen features where our pipeline recovers fifty-three — so comparing published feature counts measures annotation effort as much as biology.

We therefore annotated twelve complete *Pleurotus* mitogenomes with a single pipeline and asked: does intron content explain size; is any of the sequence duplicated; can the uncharacterised open reading frames be assigned a function or at least shown to be real; and, since mitochondrial function is jointly encoded by two genomes, is nuclear-encoded mitochondrial expression coordinated with the mitochondrial genome's own output across tissues.

2 Results

2.1 Mitogenome content

The *P. ostreatus* BOM_ss5 mitochondrion is 71,949 bp at 26.6% GC and, under the corrected annotation, carries 20 protein-coding genes, 25 tRNAs, 2 rRNAs and 6 spliced introns (Fig. 1). Coverage is markedly uneven: the ribosomal RNA regions reach 10^6 while much of the protein-coding complement sits between 10^1 and 10^3.

2.2 Intron content tracks genome size, within and between species

Across twelve mitogenomes, length spans 60,694 bp (*P. citrinopileatus*) to 102,950 bp (*P. giganteus*), a 1.7-fold range with no corresponding change in the respiratory gene set (Table 1).

Detected group I intron content correlates with length at $r = +0.89$, and ORF count more strongly at $r = +0.96$ (Fig. 2). The second correlation is not independent of the first: group I introns encode homing endonucleases, so an intron gain adds both sequence and an ORF. Together they support the standard account.

The effect operates below the species level. Among the seven *P. ostreatus* mitogenomes here, length ranges from 65,277 bp (PC9.15) to 73,085 bp (N009) — a 7.8 kb spread within one species, with the smaller genomes carrying three to four detected group I intron hits and the larger seven to eight. Our own strain sits at the larger end.

Detected intron span is a lower bound: covariance models capture the conserved core rather than the whole intron. For BOM_ss5 the models detect 1,123 bp where split-read evidence in the companion paper puts the true spliced total at 4,332 bp, roughly fourfold higher. Detected span is therefore a comparable index across genomes, not an absolute measurement, and it does not by itself account for the whole size difference.

2.3 Little duplication, but what there is duplicates an unknown gene

Self-alignment recovers only two internal repeats of at least 200 bp, together 802 bp or 1.1% of the genome. This mitogenome is not built from repeated sequence.

Both repeats are nonetheless interesting. All-versus-all comparison of the 51 predicted ORFs finds four paralogous families. Three are expected: the NADH dehydrogenase subunits nad2, nad4 and nad5 share ancestry rather than representing duplication, and two families of intron-encoded endonucleases (aI3, aI5, aI8) are homing endonucleases, which proliferate by their nature.

The fourth family is not expected. `CM148777.1_26` (64,085–64,510) and `CM148777.1_28` (66,884–67,339) have no Swiss-Prot homologue, and they correspond exactly to the larger internal repeat at 84.5% nucleotide identity — a duplicated uncharacterised ORF. Both copies are present in seven of eleven other *Pleurotus* mitogenomes, so the duplication predates the strain, but both are poorly expressed here (five reads and zero reads respectively), which is equally consistent with a pseudogenised pair.

2.4 Sensitive search does not name the uncharacterised ORFs, but conservation and expression identify four candidates

Twenty predicted ORFs have no Swiss-Prot hit. Scanning them against Pfam — a profile method substantially more sensitive than BLAST — assigns an informative domain to **none**. The single hit obtained is to a *Caenorhabditis* neuropeptide-like family at $E = 8.8 \times 10^{-4}$, which we regard as a low-complexity artefact rather than a functional assignment.

Homology search is therefore exhausted, and we fall back on two orthogonal criteria: is the ORF conserved in other mitogenomes, and is it transcribed? Four ORFs satisfy both, being present in at least six of eleven other *Pleurotus* mitogenomes and detected in at least eight of sixteen libraries (Table 2, Fig. 3a). One further ORF is present in a single other genome and is most likely strain-specific or spurious.

These four are candidate mitochondrial genes that no current database can name. We make no claim about what they do.

2.5 A predicted nuclear-encoded mitochondrial proteome

Reciprocal best hits between the 12,521 nuclear proteins and MitoCarta3.0 give 399 predicted mitochondrial proteins; a one-way screen would have returned 844, which is why reciprocity was required. Curated Swiss-Prot mitochondrial localisations independently identify 467. The union is **631 proteins, 5.0% of the proteome**.

The two approaches are complementary by construction. MitoCarta is a mammalian inventory, so fungal-specific mitochondrial proteins are invisible to it; Swiss-Prot contains curated yeast and filamentous-fungal entries but covers only 41% of this proteome at all. The union is conservative and should be read as a floor.

2.6 Mito-nuclear balance differs between tissues, but the ranges overlap

Because mitochondrial function is encoded by two genomes, their relative output is measurable. Expressing both on the same DESeq2 size factors, the ratio of mtDNA-encoded to nuclear-encoded mitochondrial expression is highest in the exudophore (median 1.51) and similar across the other three tissues (0.53–0.74; Table 3, Fig. 3b).

Both exudophore replicates exceed every fuzzy mycelium replicate. They do not exceed every library: one exuding mycelium library reaches 1.02 and one nodule library 0.94, against 1.14 for the lower exudophore replicate. With $n = 2$ for the exudophore and overlapping ranges, this is an observation to follow up rather than a result.

We flag one methodological point behind it. An earlier version of this comparison used raw mitochondrial counts against normalised nuclear counts, and reported the highest ratio in nodule with a correlation of +0.92 across tissues. Both were artefacts of the deepest library sitting in that group. Applying matched size factors moved the maximum to the exudophore and dropped the correlation to +0.43. Ratios between two genomes are only interpretable when both carry the same normalisation.

3 Discussion

The comparative result is a clean confirmation of an accepted model, obtained under a consistent annotation: within *Pleurotus*, mitogenome size tracks group I intron content and the ORF load that accompanies it, and the relationship holds within *P. ostreatus* as well as between species. Whether the intron differences between strains reflect recent mobility or long-standing lineage differences is not answerable from sequence alone.

The more suggestive observations are negative or preliminary. That Pfam names none of twenty uncharacterised ORFs, while four are both conserved and expressed, is a statement about databases rather than about the organism: mitochondrial ORFs of this kind are numerous in fungal mitogenomes [1] and remain almost entirely uncharacterised. A duplicated uncharacterised ORF conserved across seven genomes but essentially untranscribed is the kind of feature that is easy to annotate away and hard to interpret.

The mito-nuclear ratio is worth pursuing precisely because the exudophore is where the other analyses in this series also point — an oxidative, respiratorily active structure. A shifted balance between the two genomes would be consistent with that. Establishing it needs more replicates than these libraries provide.

3.1 Limitations

Annotation is computational throughout; no gene model has been verified experimentally. The comparative set is twelve genomes, adequate for a correlation but not for phylogenetically controlled inference, and the species are related enough that the points are not independent. Detected intron span underestimates true intron content roughly fourfold and does so unevenly. The predicted nuclear mito-proteome inherits MitoCarta's mammalian bias and Swiss-Prot's coverage limits. All tissue statements rest on twelve libraries with the exudophore at $n = 2$. Nothing here demonstrates retrograde signalling; the data speak to expression balance only.

4 Methods

Twelve complete *Pleurotus* mitogenomes were retrieved from NCBI and annotated with the pipeline described in the companion paper: tRNAscan-SE [2], barrnap [3], EMBOSS `getorf` under genetic code 4, and DIAMOND [4] identification against Swiss-Prot [5]. Group I and II introns were detected with Infernal [6] `cmsearch` against Rfam [7] models at $E < 0.01$. BOM_ss14, the sibling nucleus of the same dikaryon, was included as a control and returned counts identical to BOM_ss5.

Internal repeats and pairwise synteny used `nucmer` [8] (`--maxmatch --nosimplify` for self-alignment), filtering the trivial diagonal and retaining alignments of at least 200 bp. ORF paralogy used all-versus-all DIAMOND with single-linkage clustering. Pfam [9] domains were assigned with `hmmscan` [10] at $E < 10^{-3}$.

MitoCarta3.0 [11] human sequences were searched reciprocally against the nuclear proteome, retaining reciprocal best hits. Swiss-Prot subcellular-location annotations were parsed from the flatfile. Mitochondrial and nuclear expression were placed on common DESeq2 [12] size factors before any ratio was computed.

Figures

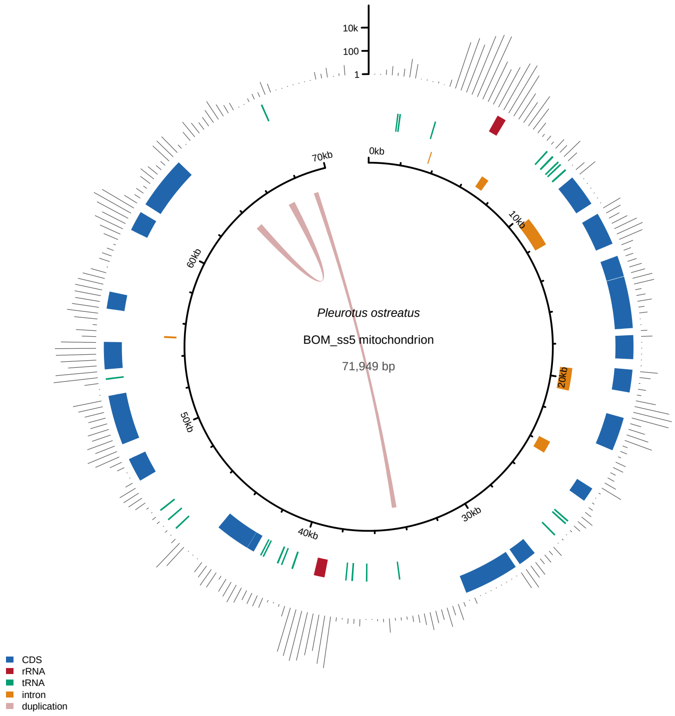

Fig. 1 The blue oyster mitochondrial genome. From the outside: pooled RNA-seq depth (log scale); protein-coding, ribosomal and transfer RNA features on the forward then reverse strand; spliced introns; coordinate axis. Interior links join the two internal repeats; the larger duplicates an uncharacterised ORF at 84.5% identity.

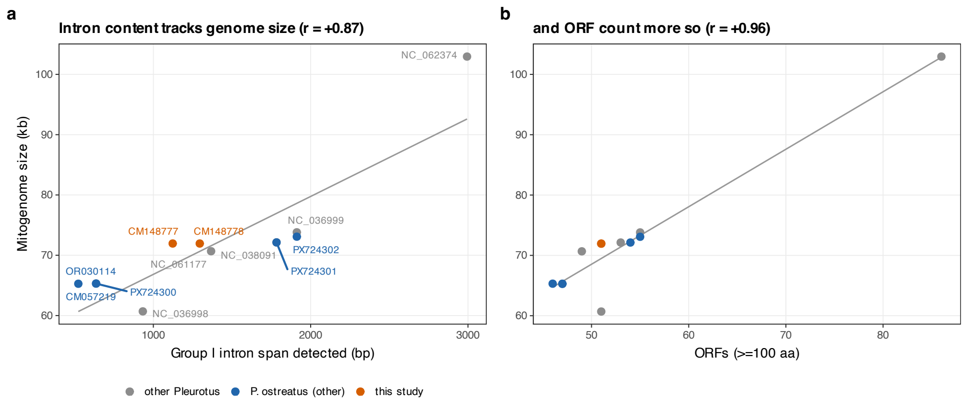

Fig. 2 Intron content tracks mitogenome size across *Pleurotus*. a, Genome size against detected group I intron span. **b**, Against ORF count. All twelve genomes annotated with one pipeline.

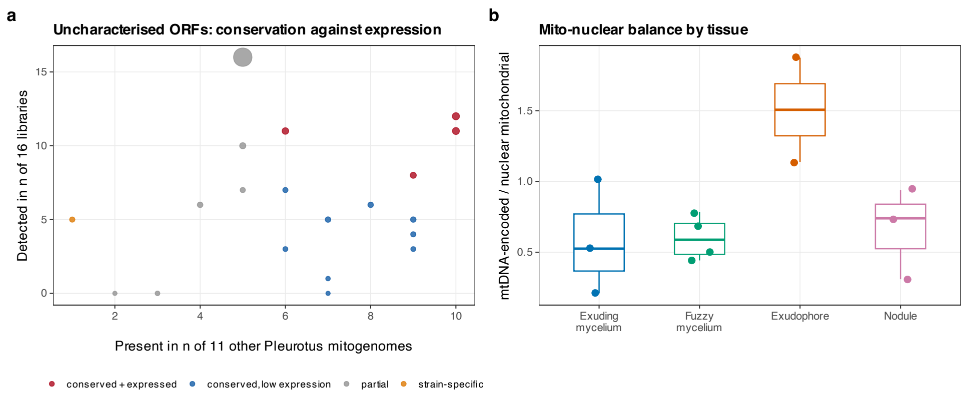

Fig. 3 Uncharacterised ORFs, and mito-nuclear balance. a, Each uncharacterised ORF by conservation across other *Pleurotus* mitogenomes and detection across libraries; point size is read support. **b**, Ratio of mtDNA-encoded to predicted nuclear-encoded mitochondrial expression, per library, both on matched size factors.

Tables

Table 1 Complete *Pleurotus* mitogenomes, all annotated with the pipeline used here rather than with their deposited features.

Species / strain	Accession	Length (bp)	GC%	tRNA	ORFs	gpI hits	Intron bp
P. citrinopileatus	NC_036998.1	60,694	29.0	24	51	2	933
P. ostreatus PC9.15	CM057219.1	65,277	26.2	25	47	3	523
P. ostreatus N001	PX724300.1	65,303	26.2	25	47	4	636
P. ostreatus	OR030114.1	65,311	26.2	25	46	4	636
P. pulmonarius	NC_061177.1	70,671	25.3	25	49	6	1,367
P. ostreatus BOM_ss14	CM148778.1	71,947	26.6	25	51	5	1,295
P. ostreatus BOM_ss5 (this study)	CM148777.1	71,949	26.6	25	51	4	1,123
P. ostreatus F515	PX724301.1	72,132	26.7	25	54	7	1,783
P. cornucopiae	NC_038091.1	72,134	26.7	25	53	7	1,783
P. ostreatus N009	PX724302.1	73,085	26.7	25	55	8	1,912
P. platypus	NC_036999.1	73,807	26.8	25	55	8	1,912
P. giganteus	NC_062374.1	102,950	25.0	25	86	12	2,994

gpI hits are Rfam group I covariance-model matches; the span they cover is a lower bound, since the models capture the conserved core rather than the full intron.

Table 2 Uncharacterised mitochondrial ORFs: conservation across *Pleurotus* against expression.

ORF	Other genomes	Libraries	Reads	In rRNA block
CM148777.1_27	10/11	12/16	584	no
CM148777.1_14	10/11	11/16	524	no
CM148777.1_44	9/11	8/16	218	no
CM148777.1_30	9/11	5/16	100	no
CM148777.1_21	9/11	4/16	60	yes
CM148777.1_37	9/11	3/16	29	yes
CM148777.1_43	8/11	6/16	106	no
CM148777.1_20	7/11	5/16	104	yes
CM148777.1_28	7/11	1/16	5	no
CM148777.1_26	7/11	0/16	0	no
CM148777.1_33	6/11	11/16	376	no
CM148777.1_51	6/11	7/16	58	yes
CM148777.1_11	6/11	3/16	30	no
CM148777.1_50	5/11	16/16	14,070	yes

No Swiss-Prot hit and no informative Pfam domain. An ORF conserved in most genomes and expressed in most libraries is a candidate gene; one present in a single strain is more likely spurious.

Data availability. https://github.com/dr-richard-barker/Myco_tissue_RNAseq. Corrected annotations for all twelve mitogenomes, the comparative inventory, the ORF conservation table and the predicted nuclear mito-proteome are included.

Author contributions. [AUTHORS: CRediT statement for all eight authors.]

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Table 3 Ratio of mtDNA-encoded to predicted nuclear-encoded mitochondrial expression, per library.

Tissue	n	Min	Median	Max
Exudophore	2	1.139	1.507	1.875
Nodule	3	0.309	0.740	0.940
Fuzzy mycelium	4	0.442	0.588	0.784
Exuding mycelium	3	0.209	0.525	1.016

Both terms carry the same DESeq2 size factors. Ranges overlap between tissues and the exudophore has $n = 2$; this is descriptive.

Competing interests. [AUTHORS: declaration.]

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